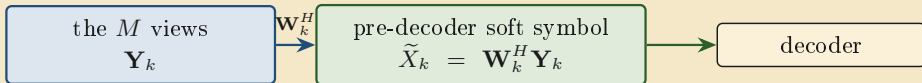


Equation (10) of juna_joint_ieee.tex

The Pilot-Trained Combiner

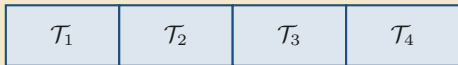
One soft symbol from the M partial-FFT views



the read-out used before decoding

Partial-FFT views

one OFDM symbol, $M = 4$ windowed pieces

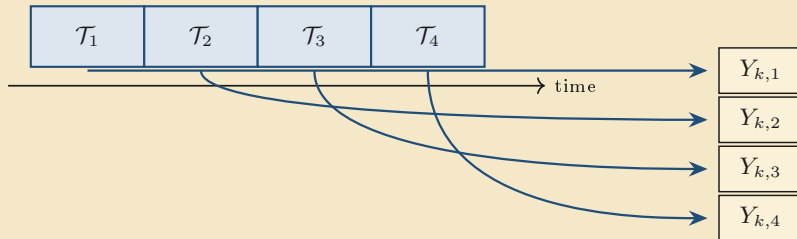


→ time

$\mathbf{Y}_k$ collects the M views of carrier k : the FFT of each windowed piece of the block, taken at bin k .

Partial-FFT views

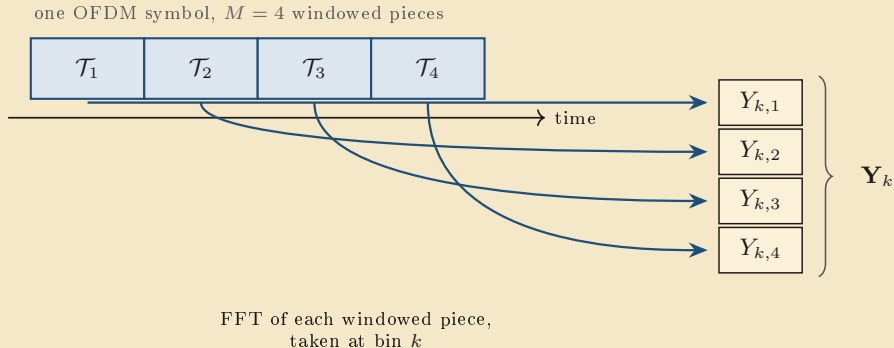
one OFDM symbol, $M = 4$ windowed pieces



FFT of each windowed piece,
taken at bin k

$\mathbf{Y}_k$ collects the M views of carrier k : the FFT of each windowed piece of the block, taken at bin k .

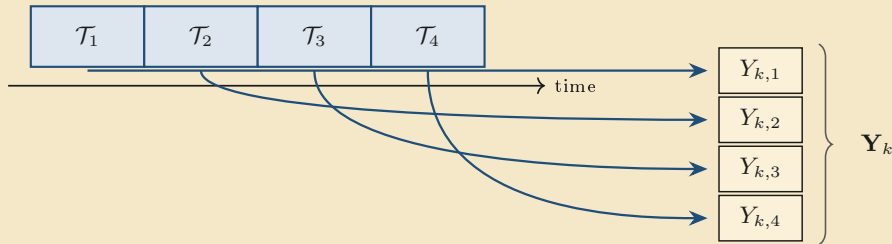
Partial-FFT views



$\mathbf{Y}_k$ collects the M views of carrier k : the FFT of each windowed piece of the block, taken at bin k .

Partial-FFT views

one OFDM symbol, $M = 4$ windowed pieces



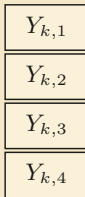
FFT of each windowed piece,
taken at bin k

$$\sum_{m=1}^M Y_{k,m} = Y_k^{\text{full}}$$

the views sum to the full-FFT bin

$\mathbf{Y}_k$ collects the M views of carrier k : the FFT of each windowed piece of the block, taken at bin k .

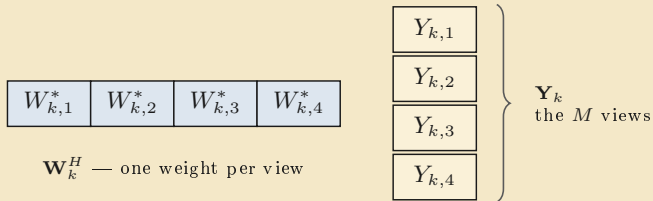
The read-out



$\mathbf{Y}_k$ — the M views

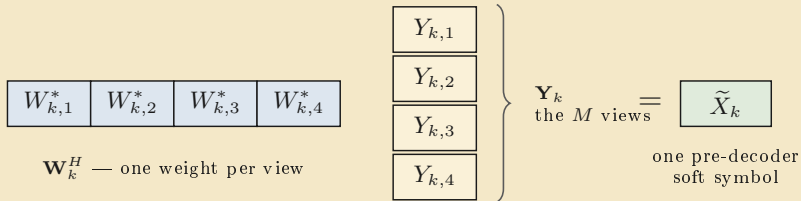
The weights rotate and scale the views back toward the transmitted symbol, and the pilots supply the carriers where that symbol is known.

The read-out



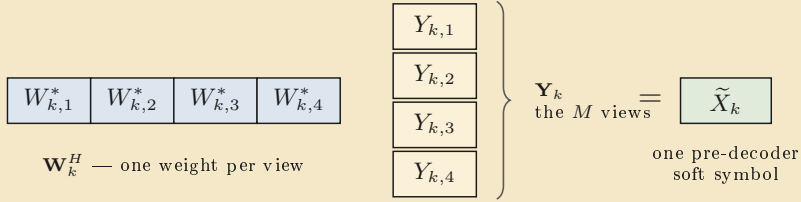
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The read-out



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The read-out



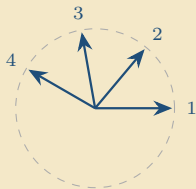
$$\tilde{X}_k = \sum_{m=1}^M W_{k,m}^* Y_{k,m}$$

$$\tilde{X}_k = \mathbf{W}_k^H \mathbf{Y}_k$$

equation (10), where $\mathbf{W}_k \in \mathbb{C}^M$ collects the combiner weights for carrier k

The weights rotate and scale the views back toward the transmitted symbol, and the pilots supply the carriers where that symbol is known.

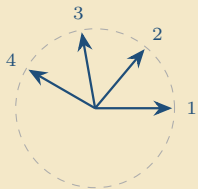
The arcs



the views of one carrier differ
by their per-segment phases

Residual Doppler fans the arcs of one carrier in phase, so their plain sum, the full-FFT scalar, is shorter than the aligned sum the views still admit.

The arcs



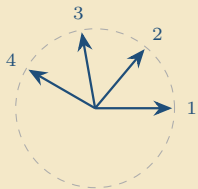
the views of one carrier differ
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plain sum:
the full-FFT scalar

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The arcs



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plain sum:
the full-FFT scalar

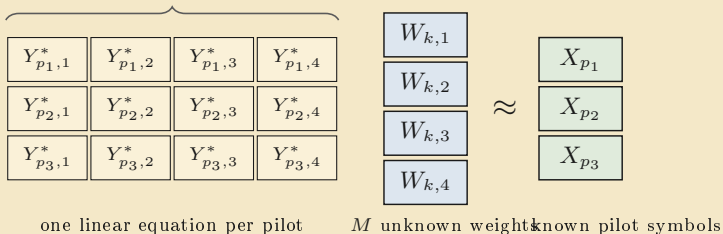


aligned sum:
one complex weight
per view realigns them

Residual Doppler fans the arcs of one carrier in phase, so their plain sum, the full-FFT scalar, is shorter than the aligned sum the views still admit.

The ridge criterion

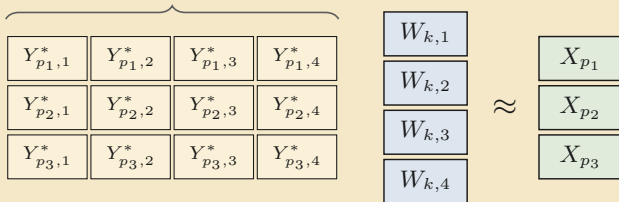
Every pilot is a carrier whose transmitted symbol the receiver knows.
the M views of each nearby pilot



The nearby pilot set has to supply at least M equations before the weights are determined, and the ridge keeps the fit stable when it supplies fewer.

The ridge criterion

Every pilot is a carrier whose transmitted symbol the receiver knows.
the M views of each nearby pilot



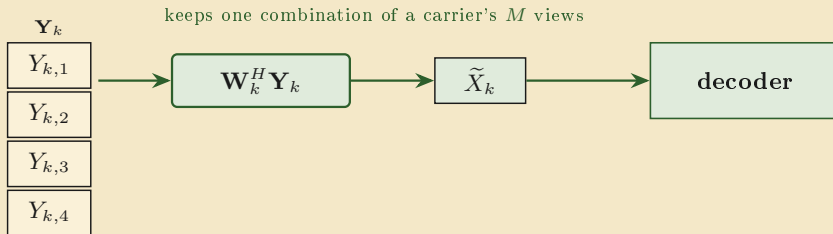
one linear equation per pilot M unknown weights M known pilot symbols

$$J_k(\mathbf{W}_k) = \sum_{p \in \mathcal{P}(k)} \left| \mathbf{W}_k^H \mathbf{Y}_p - X_p \right|^2 + \delta \|\mathbf{W}_k\|_2^2$$

equation (11), where $\mathcal{P}(k)$ is the nearby pilot set
for carrier k and $\delta > 0$ is the ridge parameter

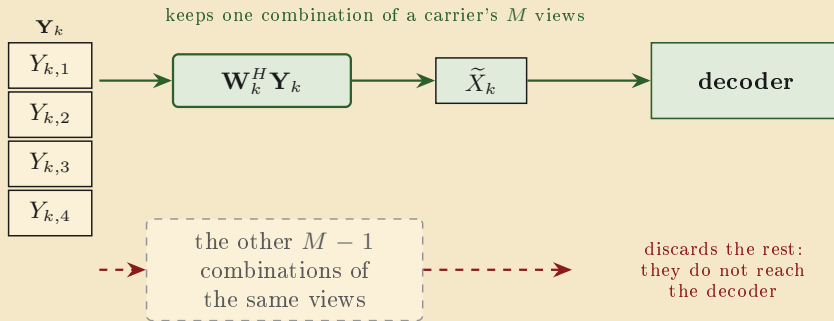
The nearby pilot set has to supply at least M equations before the weights are determined, and the ridge keeps the fit stable when it supplies fewer.

The discard



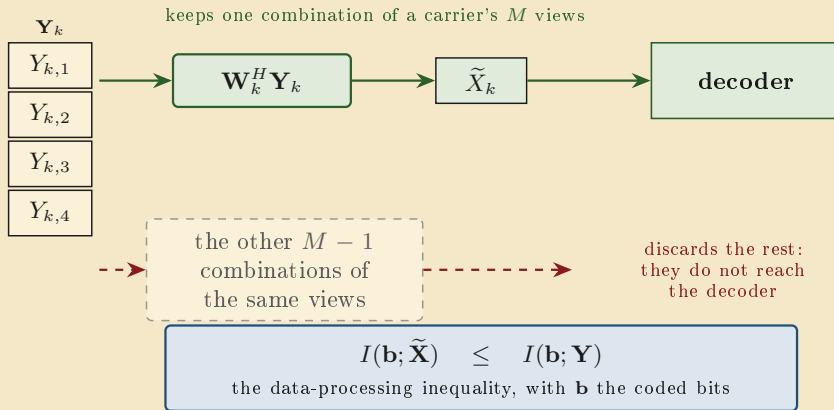
Whatever the discarded combinations say about neighbouring coded symbols — each tied by the parity constraints to every other coded bit — does not reach the decoder.

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The pilot-trained combiner

$$\tilde{X}_k = \mathbf{W}_k^H \mathbf{Y}_k$$

one weight per view; one soft symbol per carrier

it keeps one combination of the M views and discards the rest